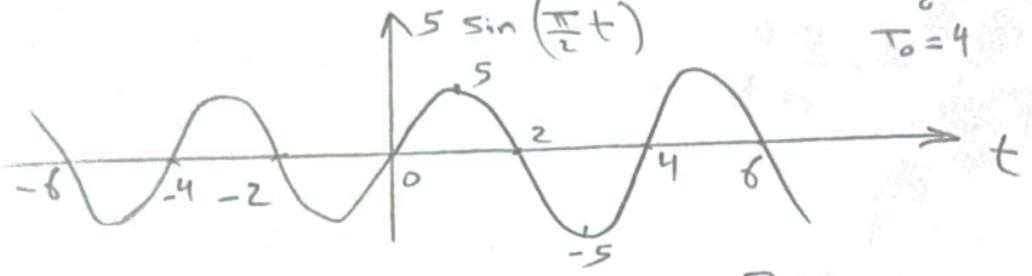
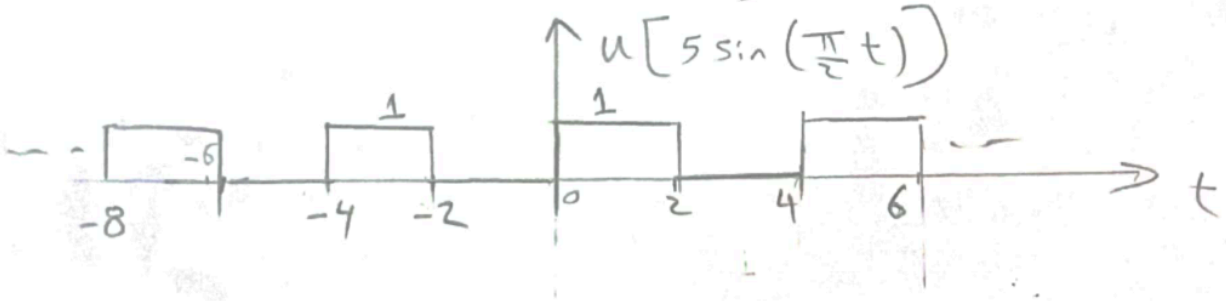
Sheet (2) Model Answer

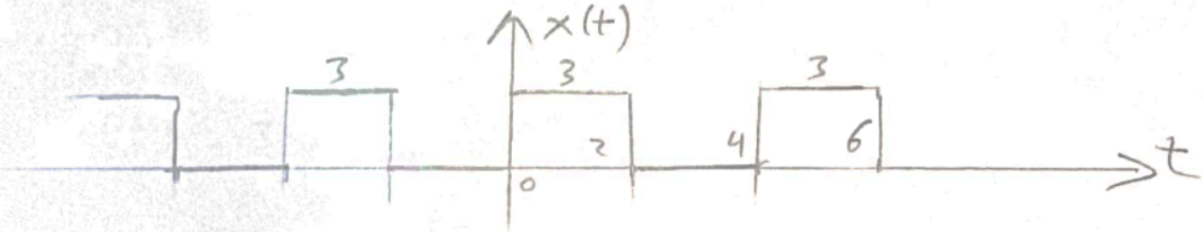
Q1) a) $x(t) = 3u\left[5\sin\left(\frac{\pi}{2}t\right)\right]$

$\omega_0 = \frac{2\pi}{T_0} = \frac{\pi}{2}$

$T_0 = 4$





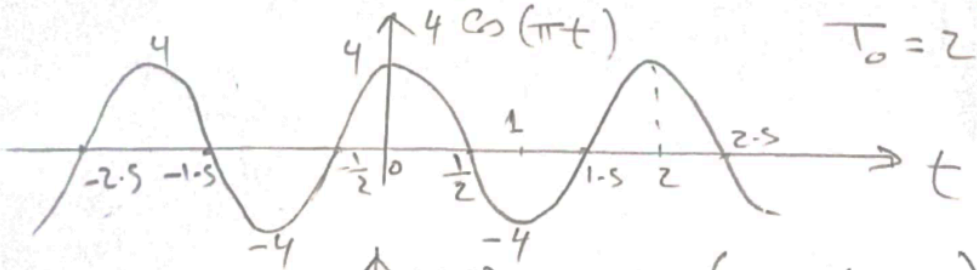


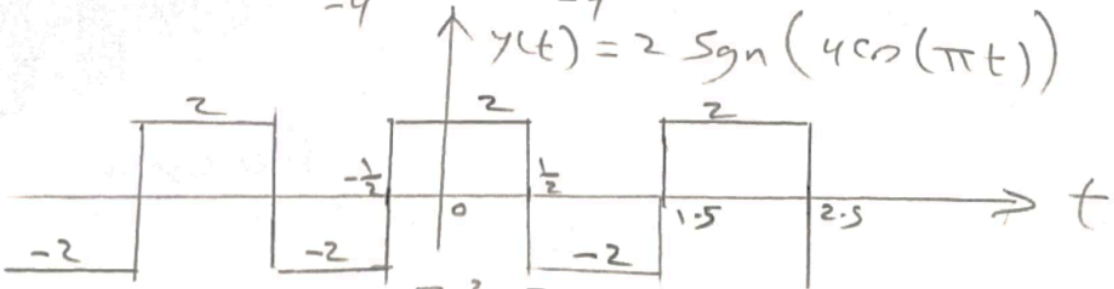
$$P_x = \frac{E_{one\ period}}{T_0} = \frac{[3^2 * 2]}{4} = 4.5\ watt$$

b) $y(t) = 2\ sgn[4\cos(\pi t)]$

$\omega_0 = \pi = \frac{2\pi}{T_0}$

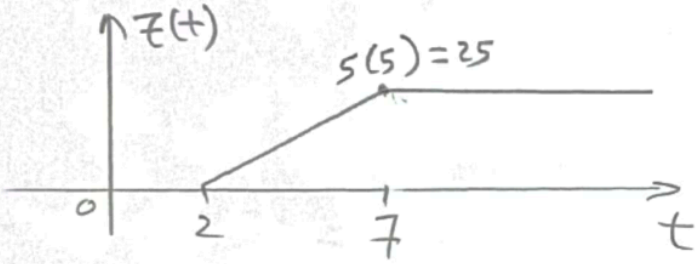
$T_0 = 2$

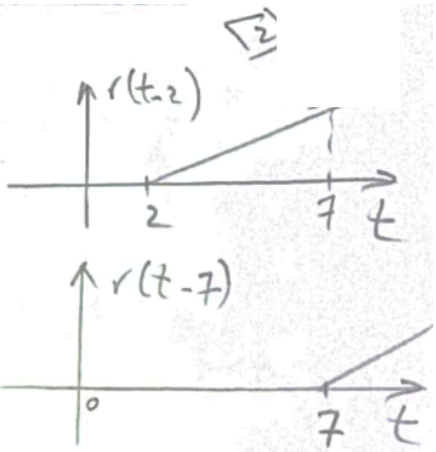




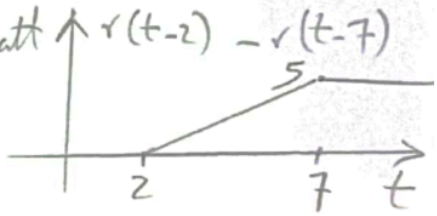
$$P_y = \frac{E_{one\ period}}{T_0} = \frac{2[2^2 * 1]}{2} = 4\ Watt$$

c) $z(t) = 5\left[r(t-2) - r(t-7)\right]$





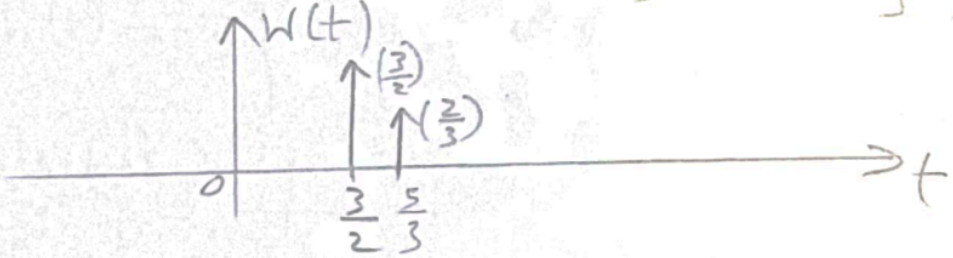
$P_z = P_u\left((25)^2\right) = \frac{1}{2}(625) = 312.5$ Watt



d) $w(t) = 3\delta(3-2t) + 2\delta(3t-5)$

$w(t) = 3\delta\left[-2(t-\frac{3}{2})\right] + 2\delta\left[3(t-\frac{5}{3})\right]$

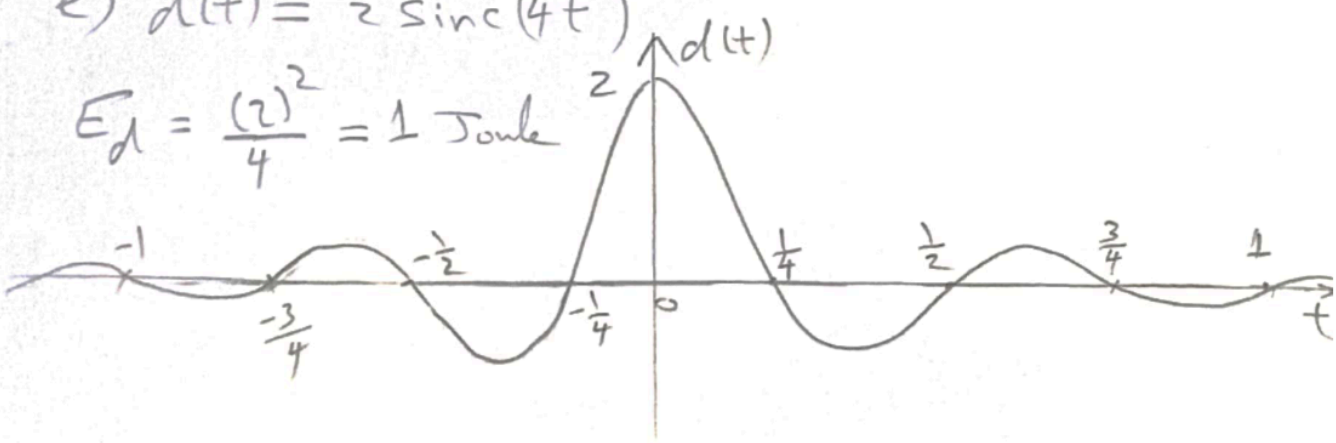
$w(t) = \frac{3}{2}\delta(t-\frac{3}{2}) + \frac{2}{3}\delta(t-\frac{5}{3})$



it is neither energy nor power signal

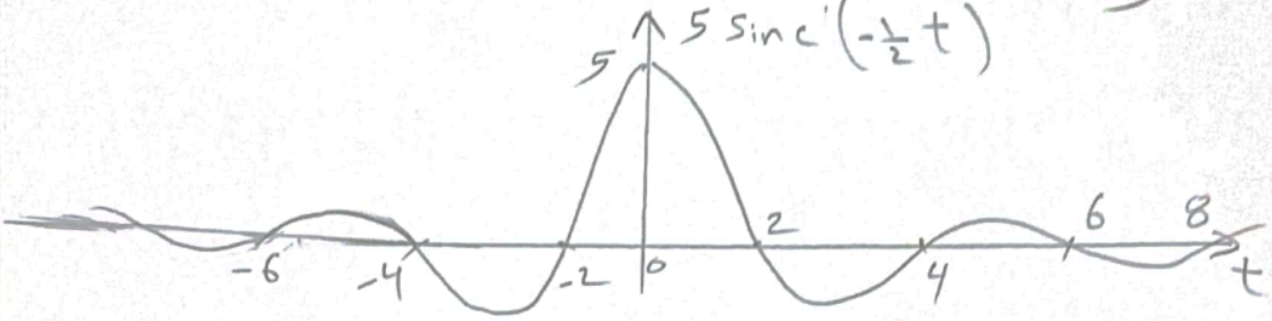
e) $d(t) = 2\,\mathrm{sinc}(4t)$

$E_d = \frac{(2)^2}{4} = 1$ Joule



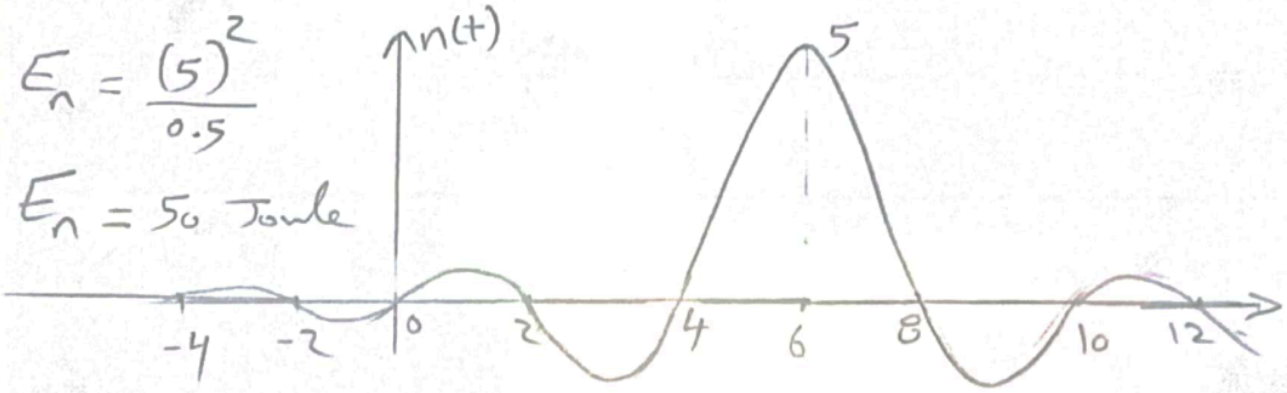
f) $n(t) = 5\,\mathrm{Sinc}\left(3-\frac{t}{2}\right) = 5\,\mathrm{sinc}\left[-\frac{1}{2}(t-6)\right]$

$5\,\mathrm{Sinc}'\left(-\frac{1}{2}t\right)$



$E_n = \frac{(5)^2}{0.5}$

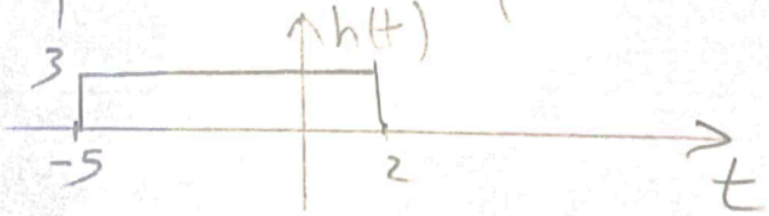
$E_n = 50$ Joule



g) $h(t) = 3\{u(t+5) - u(t-2)\}$

$E_h = (3)^2(7)$

$E_h = 63$ Joule



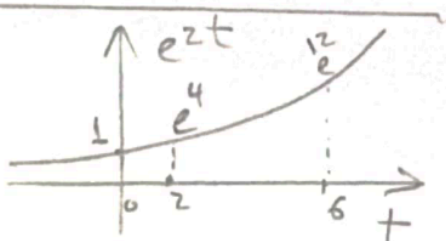
h) $k(t) = e^{2t}\{u(t-2) - u(t-6)\}$

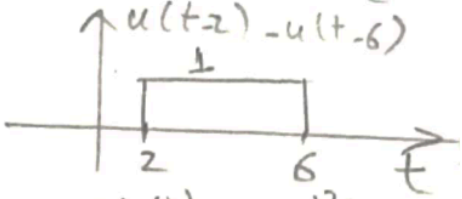
$E_k = \int_2^6 (e^{2t})^2 dt$

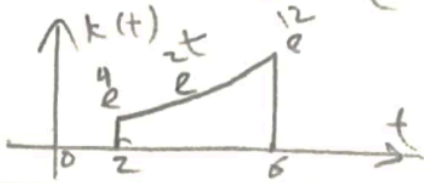
$E_k = \int_2^6 e^{4t} dt$

$E_k = \frac{1}{4}\left[e^{4t}\right]_2^6$

$E_k = \frac{1}{4}\left\{e^{24} - e^{8}\right\} = 6.62 \times 10^9$ Joule





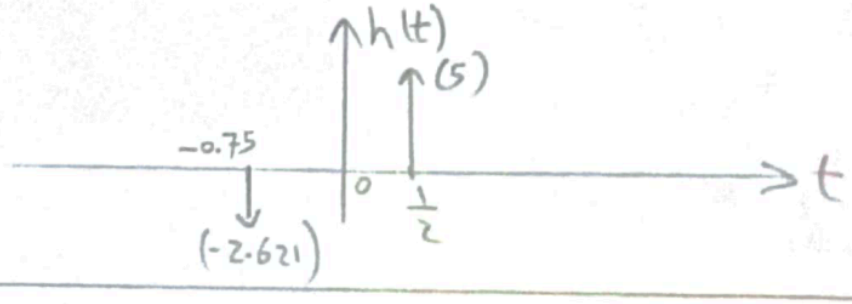


Q2

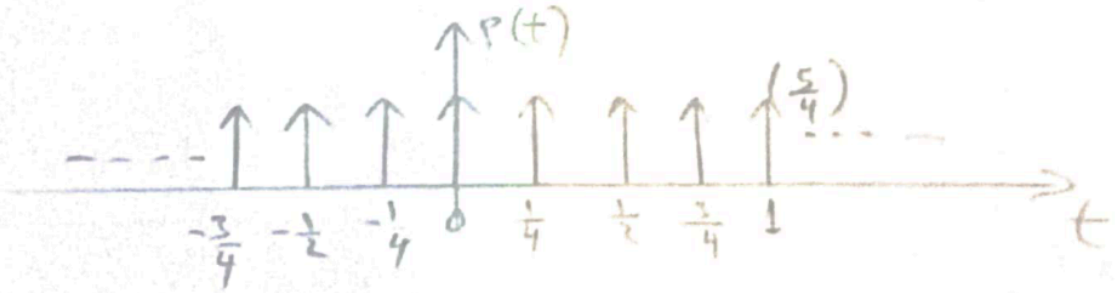
a) $h(t) = \{2t + 3\sin(\pi t) + 1\}\{\delta(t-\frac{1}{2}) + \delta(t+0.75)\}$

$h(t) = \{2(\frac{1}{2}) + 3\sin(\frac{\pi}{2}) + 1\}\delta(t-\frac{1}{2}) + \{2(-\frac{3}{4}) + 3\sin(-\frac{3}{4}\pi) + 1\}\delta(t+0.75)$

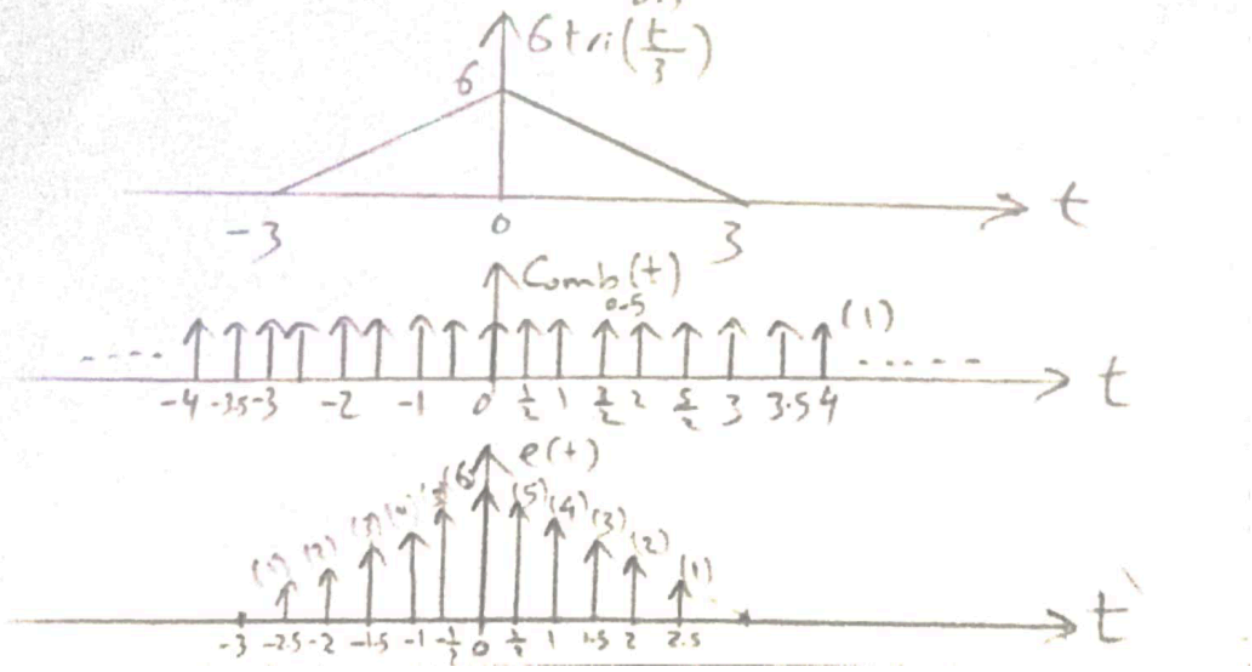
$h(t) = 5\,\delta(t-\frac{1}{2}) - 2.621\,\delta(t+0.75)$



b) $p(t) = 5\,Comb(4t)$



c) $e(t) = 6\,tri(\frac{t}{3}) \cdot Comb_{0.5}(t)$

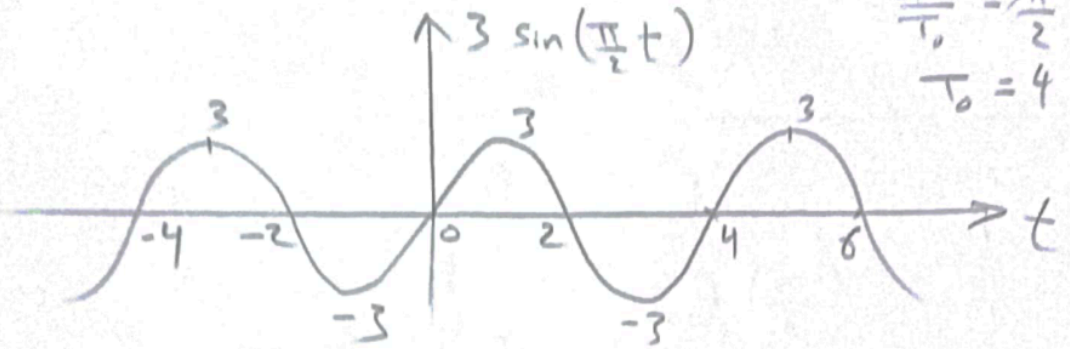


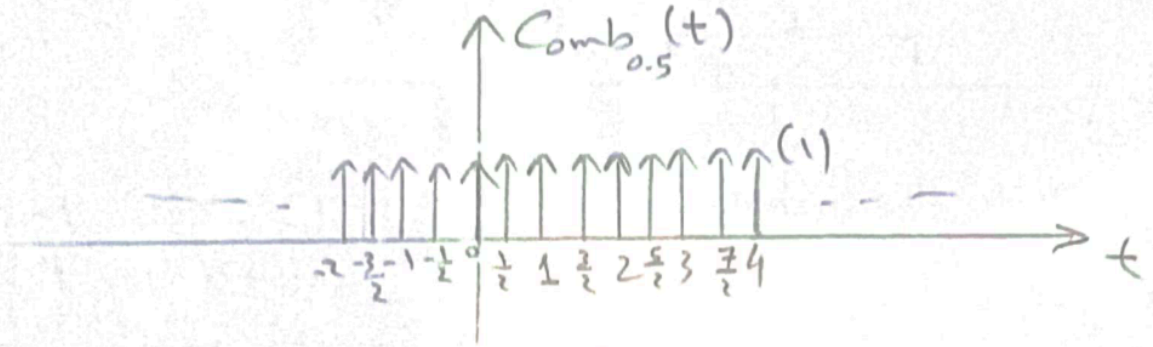
d) $s(t) = 3\sin\left(\frac{\pi}{2}t\right) \cdot Comb_{0.5}(t)$

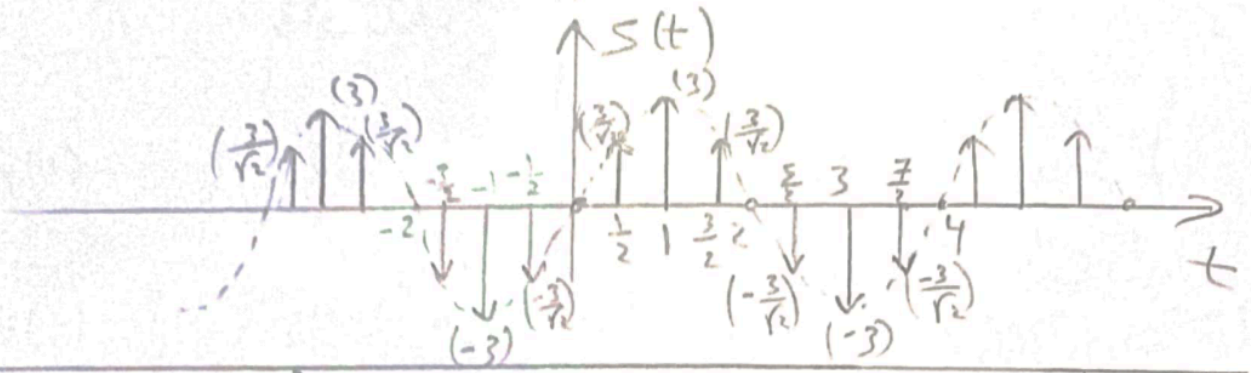
$\omega_0 = \frac{\pi}{2}$

$\frac{2\pi}{T_0} = \frac{\pi}{2}$

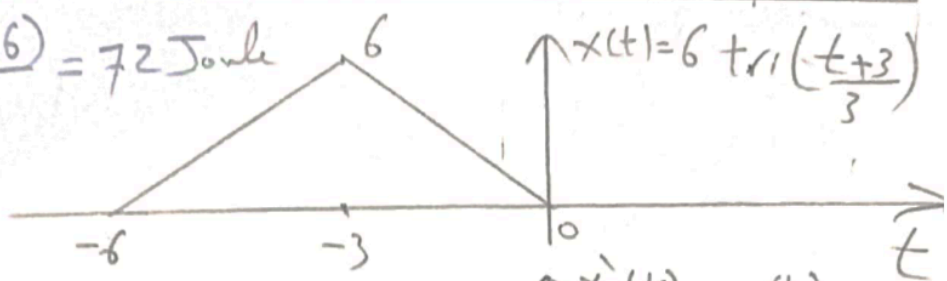
$T_0 = 4$





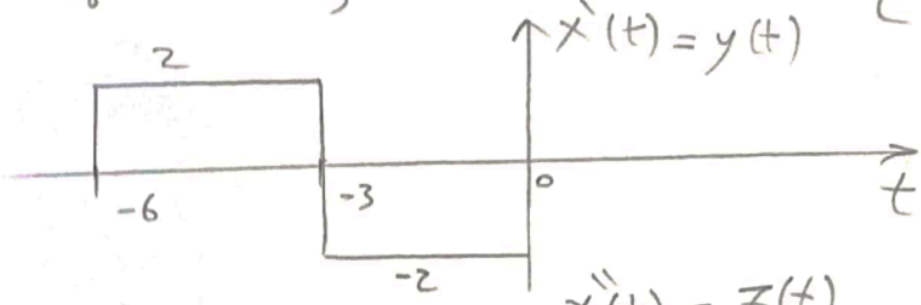


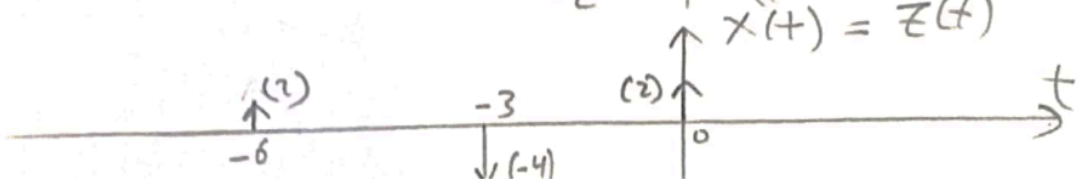
Q3] $E_x = \frac{(6)^2(6)}{3} = 72$ Joule



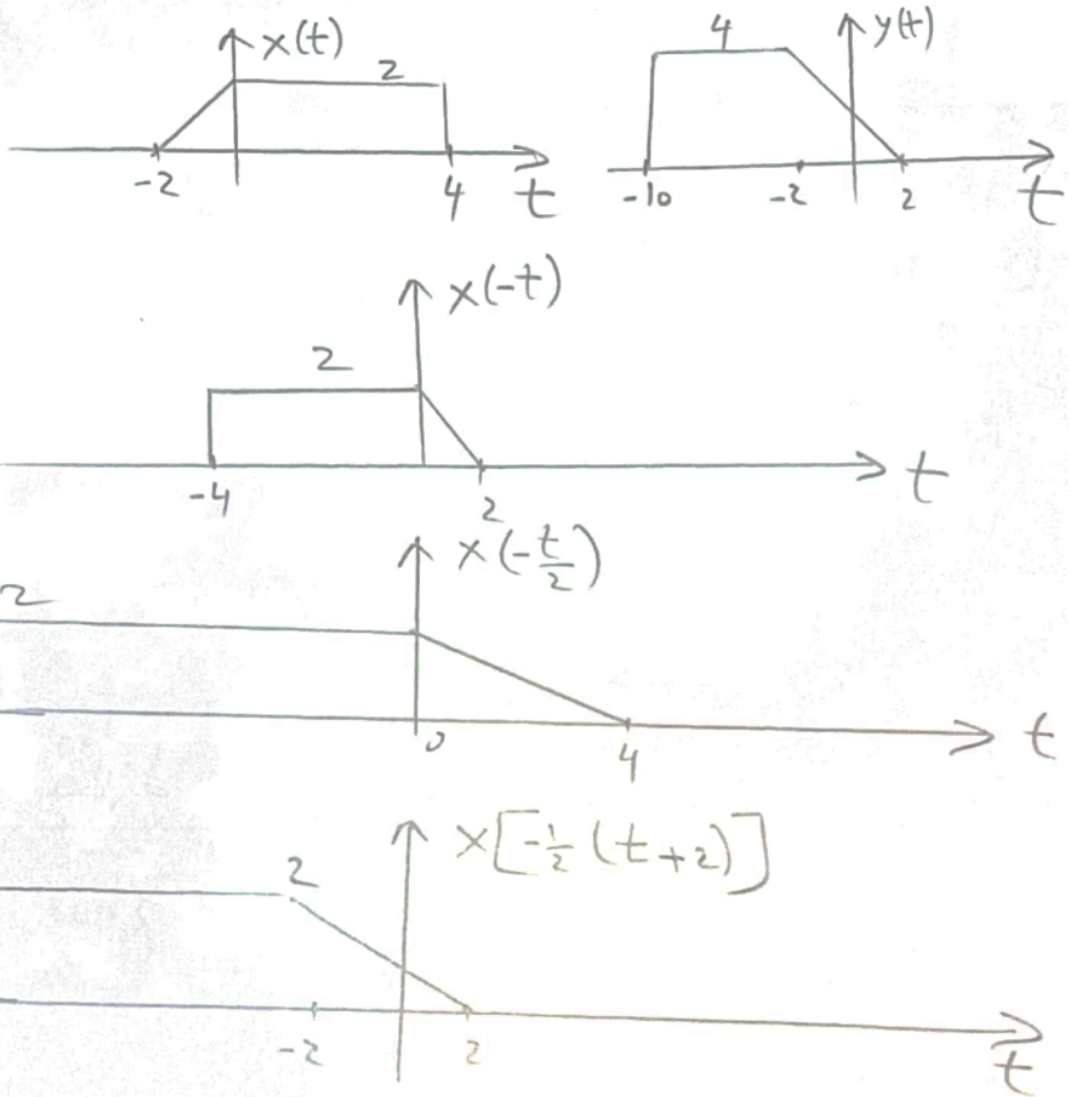
$E_y = 2\left((2)^2(3)\right)$

$E_y = 24$ Joule





Q4



$\therefore y(t) = 2 \times \left[-\frac{1}{2}(t+2)\right]$

$y(t) = 2 \times \left(-\frac{1}{2}t - 1\right)$

Q5 (a) $\int_{-\infty}^{\infty} \left[7t^3 - 2\sin\left(\frac{\pi}{8}t - \frac{\pi}{3}\right) + 2\right]\left[\delta(t-2) + \delta(-3t-6)\right] dt$

$\delta(-3t-6) \rightarrow \delta[-3(t+2)] \rightarrow \frac{1}{3}\delta(t+2)$

$I = \left[7(2)^3 - 2\sin\left(\frac{\pi}{4} - \frac{\pi}{3}\right) + 2\right] \int_{-\infty}^{\infty} \delta(t-2)\,dt \rightarrow 1$

$+ \frac{1}{3}\left[7(-2)^3 - 2\sin\left(-\frac{\pi}{4} - \frac{\pi}{3}\right) + 2\right] \int_{-\infty}^{\infty} \delta(t+2)\,dt \rightarrow 1$

$$I = 58.518(1) - 17.356(1) = 41.162$$

(b) $$\int_{-2}^{10} \left(e^{-2t} + 2t + 5\right)\left[4\delta(2t+3) + \delta(2t+10) + \delta(t-12)\right]dt$$

$\delta(2t+3) \rightarrow \delta[-2(t-\frac{3}{2})] \rightarrow \frac{1}{2}\delta(t-\frac{3}{2})$

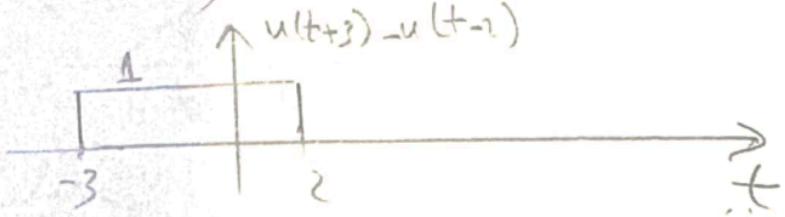
$\delta(2t+10) \rightarrow \delta[2(t+5)] \rightarrow \frac{1}{2}\delta(t+5)$

$$I = \frac{4}{2}\left[e^{-2(\frac{3}{2})} + 2(\frac{3}{2}) + 5\right]\int_{-2}^{10} \delta(t-\tfrac{3}{2})\,dt + 0 + 0$$

$\int \delta(t-\frac{3}{2})\,dt \rightarrow 1$

$$I = 16.0996$$

(c) $$\int_{-\infty}^{t} \left[\delta(t+3) - \delta(t-2)\right]dt = u(t+3) - u(t-2)$$



(d) $$\int_{-5}^{5} \underbrace{[5t^3}_{\text{odd}} + \underbrace{2t}_{\text{odd}} + \underbrace{6}_{\text{even}}]\,\underbrace{(\cos\pi t)}_{\text{even}}\,dt$$

$$\int_{-5}^{5} \underbrace{(5t^3 \cdot \cos\pi t)}_{\text{odd}}\,dt + \int_{-5}^{5} \underbrace{(2t)(\cos\pi t)}_{\text{odd}}\,dt + 6\int_{-5}^{5} \underbrace{\cos(\pi t)}_{\text{even}}\,dt$$

$\rightarrow 0$ $\rightarrow 0$

$$I = 0 + 0 + 6(2)\int_{0}^{5} \cos(\pi t)\,dt = \frac{12}{\pi}\left[\sin\pi t\right]_{0}^{5}$$

$$I = 0$$

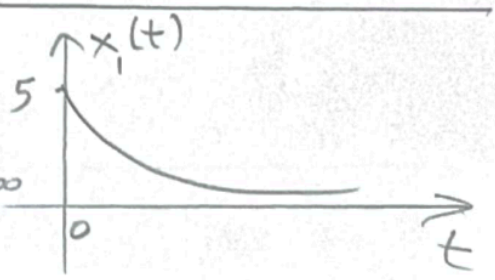
Q6

(a) $x_1(t) = 5e^{-2t}u(t)$

$E_{x_1} = \int_0^\infty (5e^{-2t})^2 dt$

$E_{x_1} = 25\int_0^\infty e^{-4t} dt = \frac{25}{-4}\left[e^{-4t}\right]_0^\infty$

$E_{x_1} = \frac{25}{-4}[0-1] = \frac{25}{4} = 6.25$ Joule.

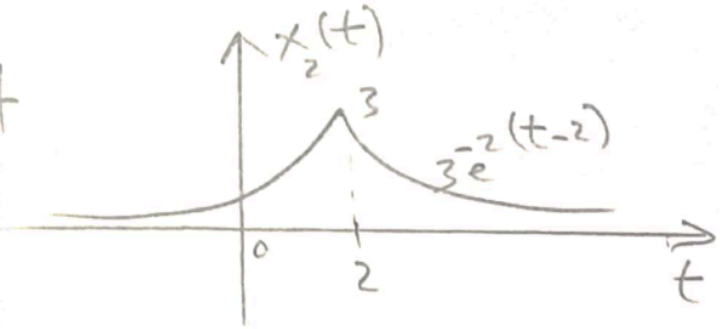


(b) $x_2(t) = 3e^{-2|t-2|}$

$E_{x_2} = 2\int_2^\infty \left[3e^{-2(t-2)}\right]^2 dt$

$E_{x_2} = 2(9)\int_2^\infty e^{-4(t-2)} dt$

$E_{x_2} = \frac{18}{-4}\left[e^{-4(t-2)}\right]_2^\infty = -4.5[0-1] = 4.5$ J

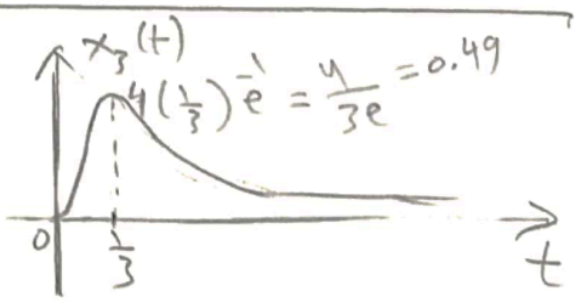


(c) $x_3(t) = 4te^{-3t}u(t)$

$E_x = \int_0^\infty 16t^2 e^{-6t} dt$

$E_x = 16\int_0^\infty t^2 e^{-6t} dt$

Note: $\int_0^\infty t^n e^{-\alpha t} dt = \frac{n!}{\alpha^{n+1}}$



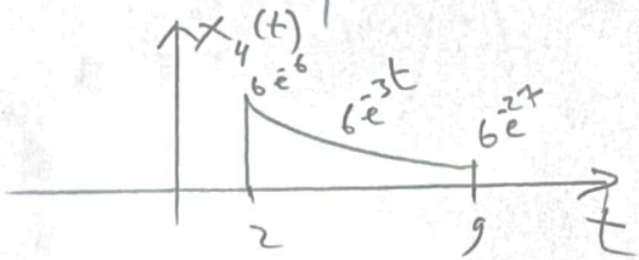
$$\therefore E_x = 16\left(\frac{2!}{6^3}\right) = \frac{16 \times 2}{216} = \frac{4}{27}$$

(d) $x_4(t) = 6\,e^{-3t}\{u(t-2) - u(t-9)\}$

$$E_x = \int_2^9 36\,e^{-6t}\,dt$$

$$E_x = \frac{36}{-6}\left[e^{-6t}\right]_2^9$$

$$E_x = -6\{e^{-54} - e^{-12}\} = 36.87\ \mu J$$

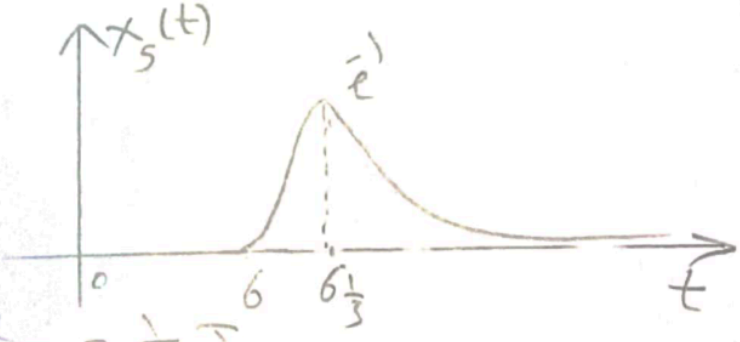


e) $x_5(t) = 3(t-6)\,e^{-3(t-6)}\,u(t-6)$

$$E_{x_5} = \int_0^\infty (3te^{-3t})^2\,dt$$

$$E_{x_5} = \int_0^\infty 9t^2 e^{-6t}\,dt$$

$$E_{x_5} = 9\left(\frac{2!}{6^3}\right) = \frac{9 \times 2}{216} = \frac{1}{12}\,J$$



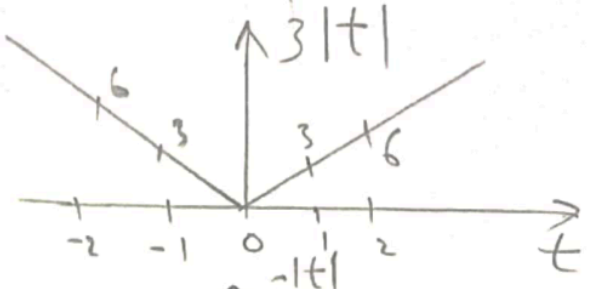
f) $x_6(t) = 3|t|\,e^{-|t|}$

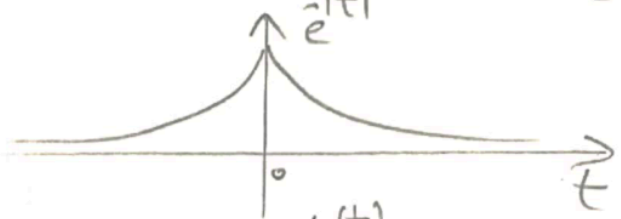
$$E_{x_6} = 2\int_0^\infty (3te^{-t})^2\,dt$$

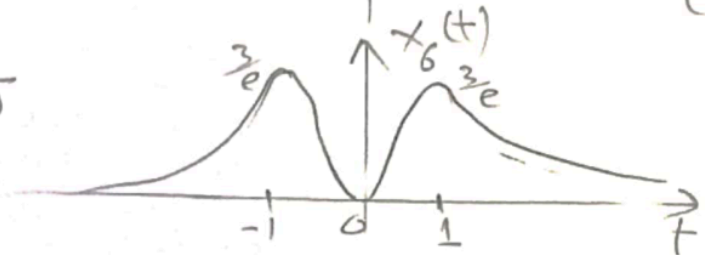
$$E_{x_6} = 2(9)\int_0^\infty t^2 e^{-2t}\,dt$$

$$E_{x_6} = 18\left(\frac{2!}{2^3}\right)$$

$$E_{x_6} = \frac{18 \times 2}{8} = 4.5\ J$$



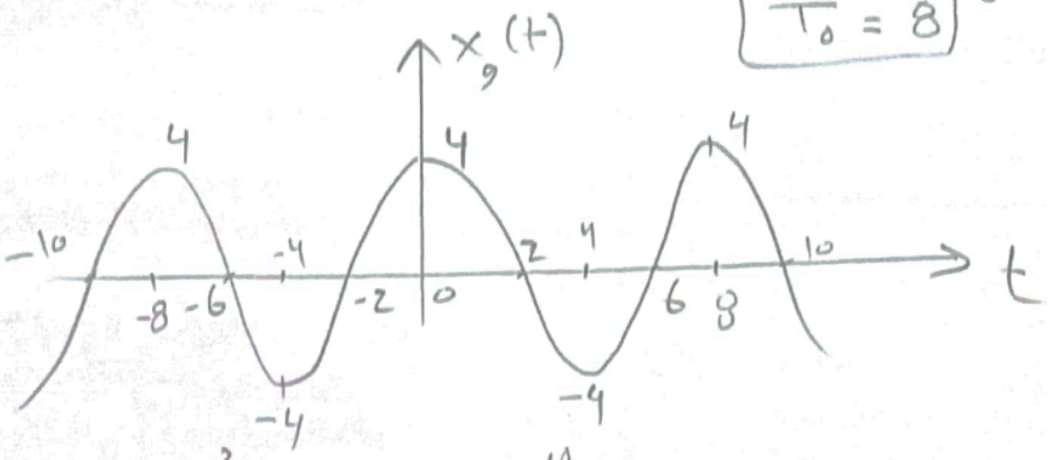




(g) $x_9(t) = 4\cos\left(\frac{\pi}{4}t\right)$

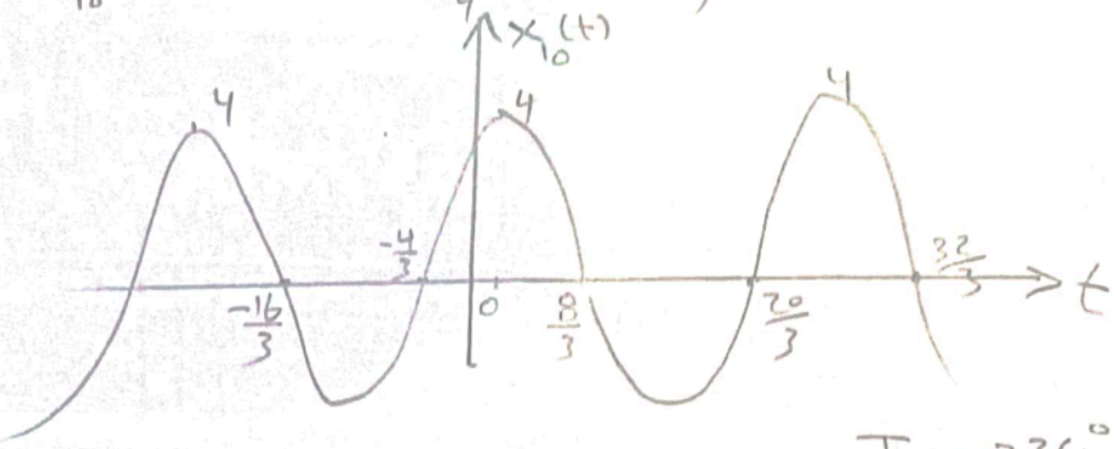
$\omega_0 = \frac{\pi}{4} = \frac{2\pi}{T_0}$

$T_0 = 8$



$P_{x_9} = \frac{(4)^2}{2} = 8$ watt

(h) $x_{10}(t) = 4\cos\left(\frac{\pi}{4}t - 30^\circ\right)$



$P_{x_{10}} = \frac{(4)^2}{2} = 8$ watt

$T_0 \longrightarrow 360^\circ$

$? \longrightarrow 120^\circ$

$? = \frac{120\,T_0}{360}$

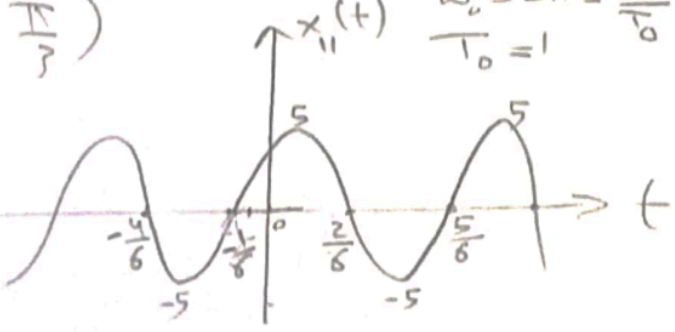
$= \frac{T_0}{3} = \frac{8}{3}$

(i) $x_{11}(t) = 5\sin\left(2\pi t + \frac{\pi}{3}\right)$

$P_{x_{11}} = \frac{(5)^2}{2} = 12.5$ Watt

$\omega_0 = 2\pi = \frac{2\pi}{T_0}$

$T_0 = 1$



$T_0 \longrightarrow 360^\circ$

$? \longrightarrow 120^\circ$

$? = \frac{T_0}{3} = \frac{1}{3} = \frac{2}{6}$